

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DevOps Lab Experiment – 10

Creating Build Pipelines: Building a Maven/Gradle Project with Azure Pipelines, Integrating Code Repositories (e.g., GitHub, Azure Repos), Running Unit Tests and Generating Reports

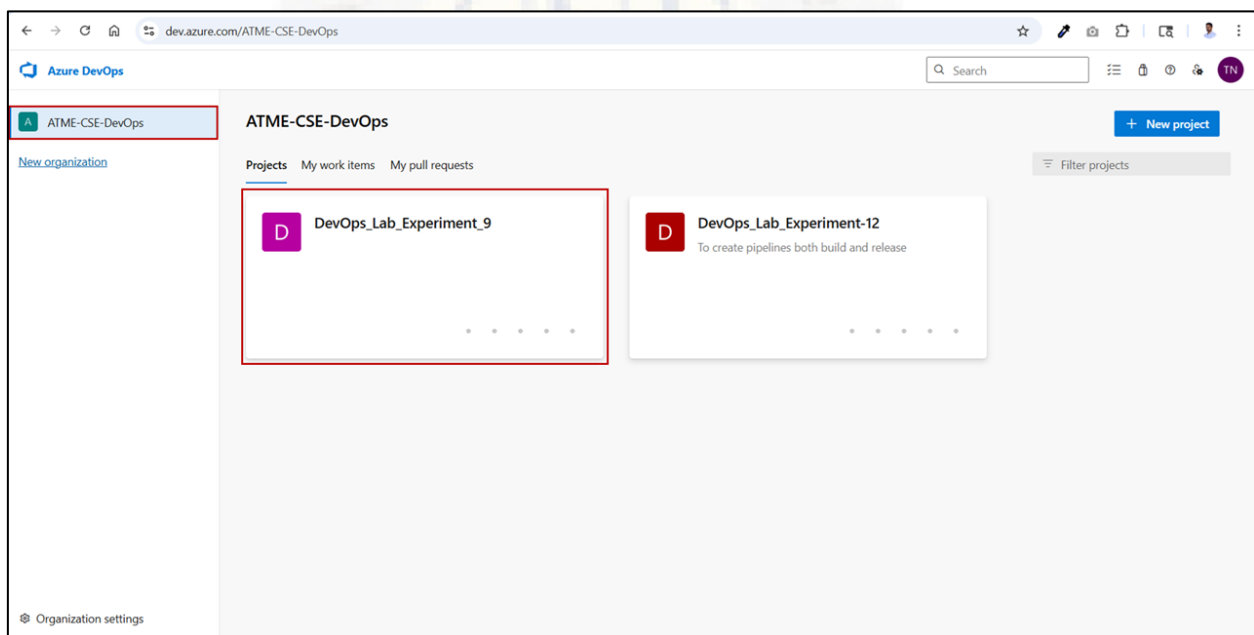
Step 1: After logging into **Azure DevOps**, select your organization name (top-left) and choose the project.

As shown below, the selected organization and project are

Organization : ATME-CSE-DevOps

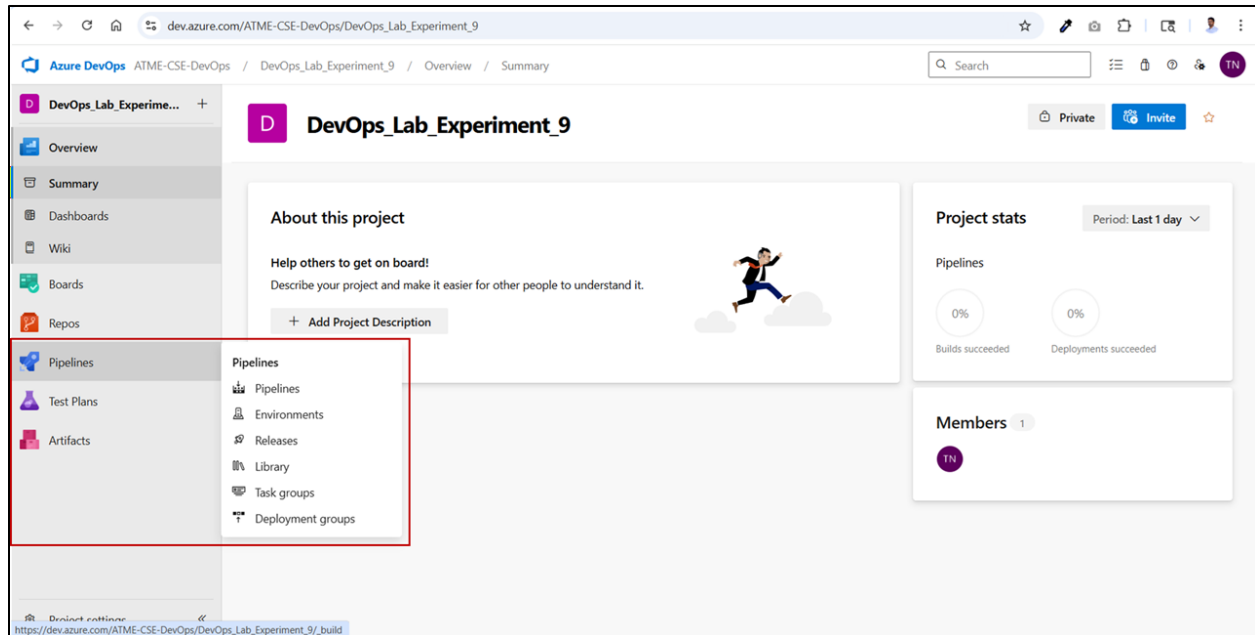
Project : DevOps_Lab_Experiment_9.

This is where all the pipeline and linked services will reside.



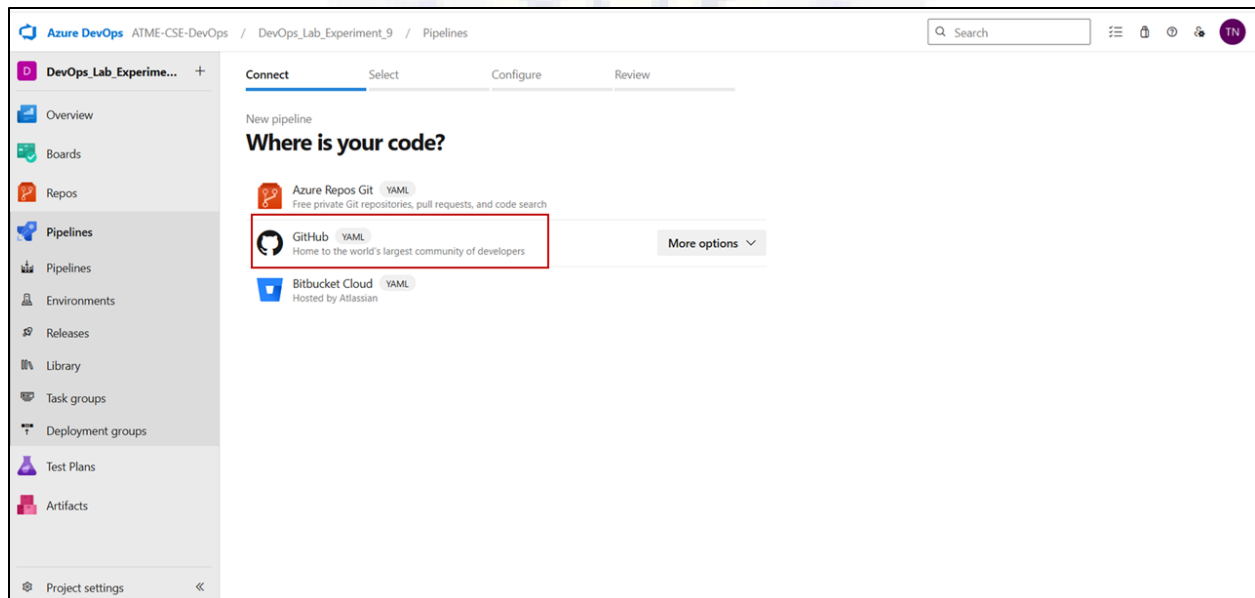
Step 2 : Inside the project, navigate to **Pipelines** → **Pipelines** and click **Create Pipeline** to start the CI/CD setup wizard.

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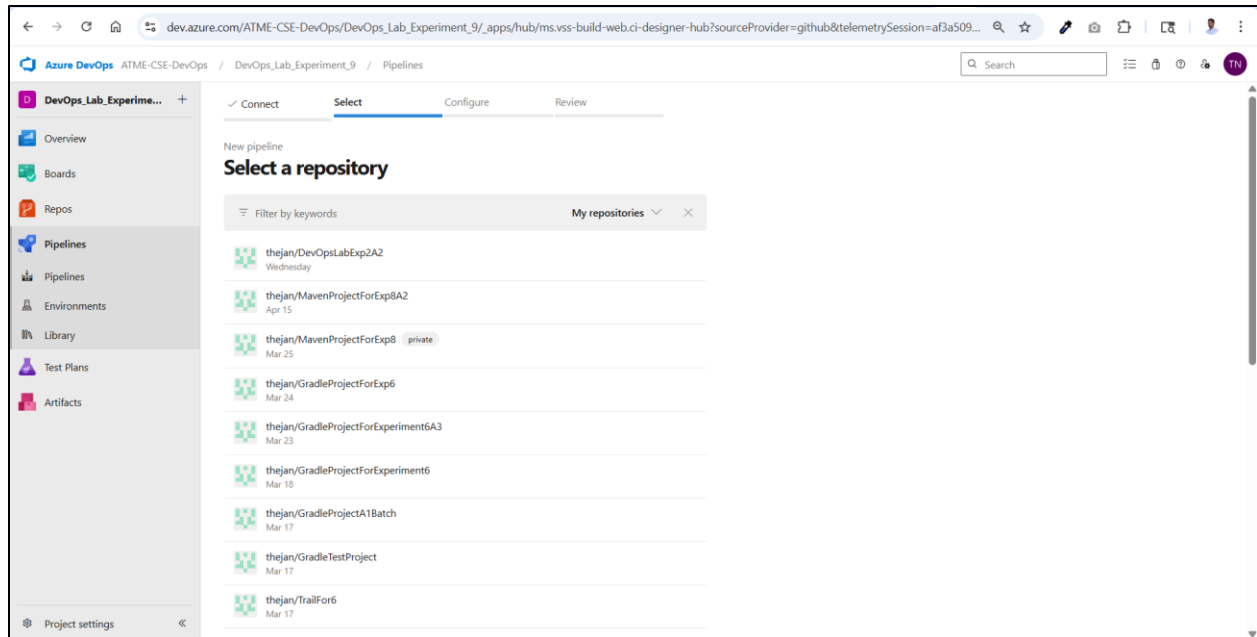
Step 3 : Connect to Your GitHub Repository

- Select GitHub and you will be redirected to authenticate and grant access.

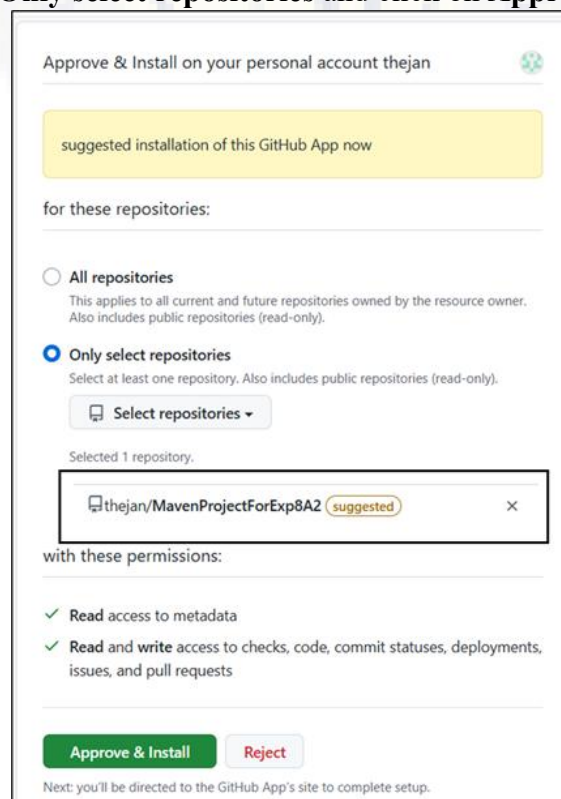


- Select a **Repository**

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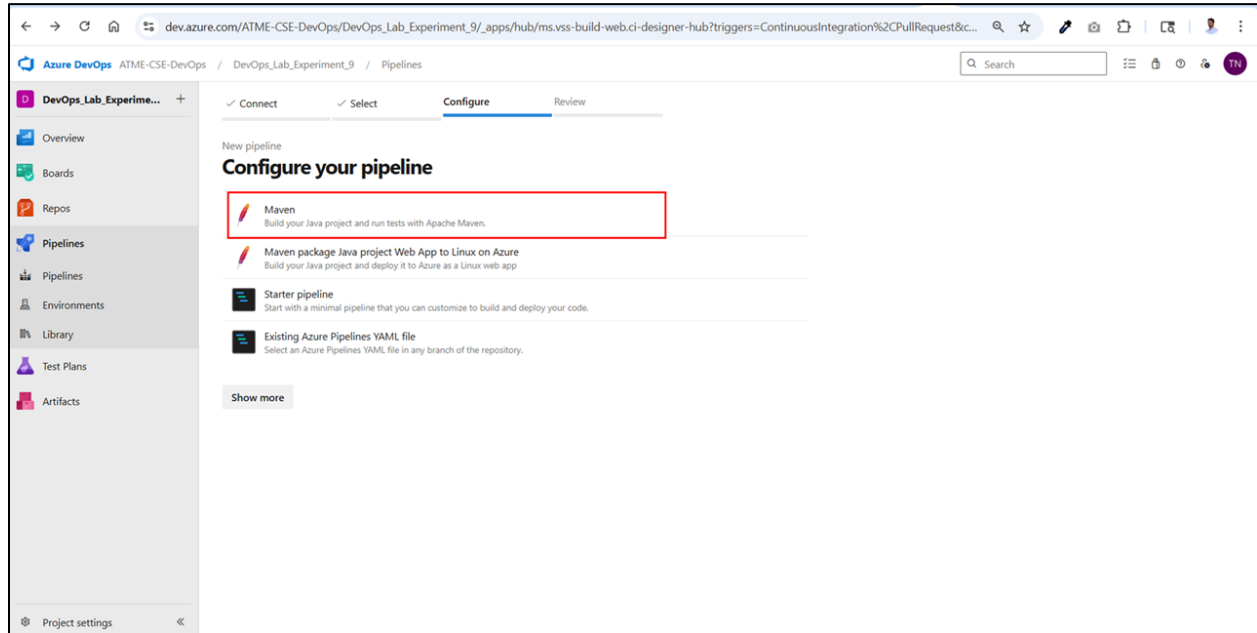


- **Approve GitHub Integration:** Approve access for Azure Pipelines to your GitHub repository. Select **Only select repositories** and click on **Approve and Install**.

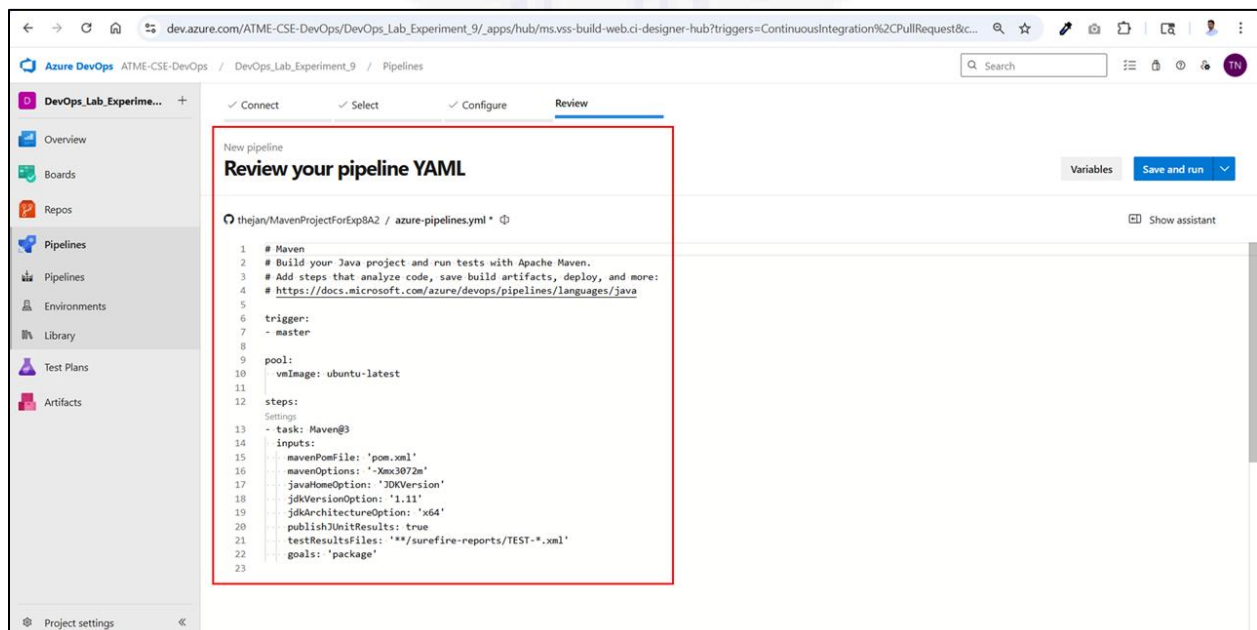


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Step 4 : Select Maven Pipeline Template: When prompted to configure the pipeline, choose **Maven** to auto-generate a pipeline template suitable for building and testing Maven project.

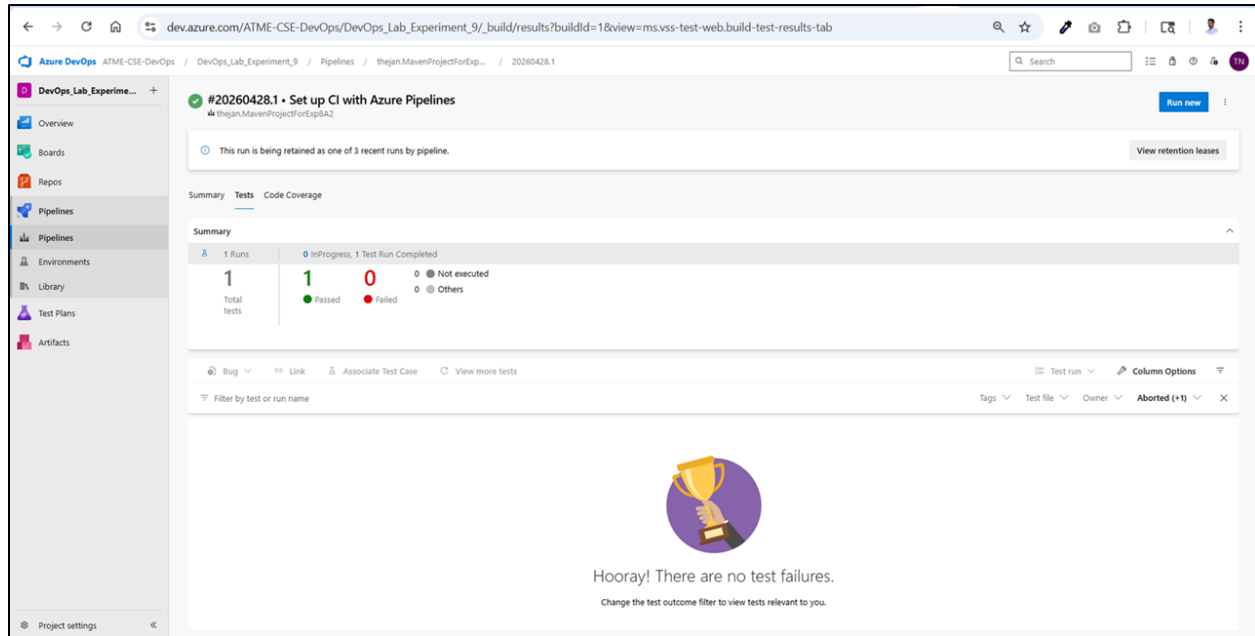


Step 5: Review and Customize YAML: Review the auto-generated azure-pipelines.yml.



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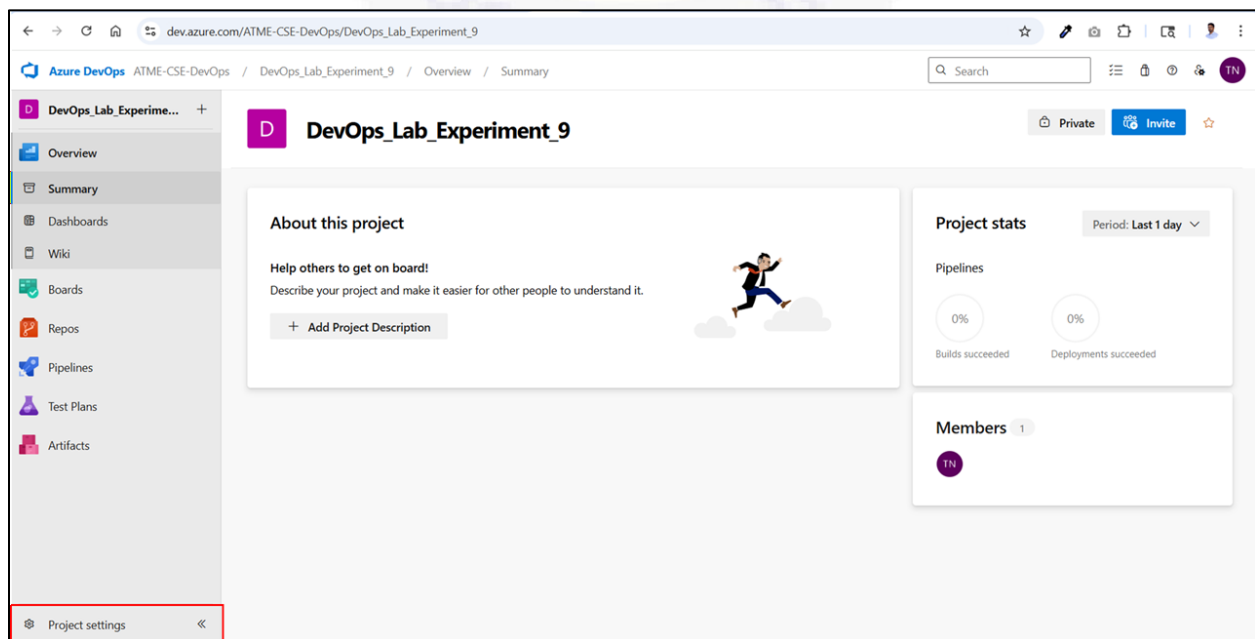
Step 6: Click Save and Run to trigger the pipeline.



The screenshot shows the Azure DevOps interface for a pipeline run. The pipeline is named "#20260428.1 • Set up CI with Azure Pipelines". The summary shows 1 total test, 1 passed, and 0 failed. A trophy icon is displayed with the message "Hooray! There are no test failures." The left sidebar shows the project settings menu.

Step 7 : Create an Agent, Configure and Run the Agent on the local machine.

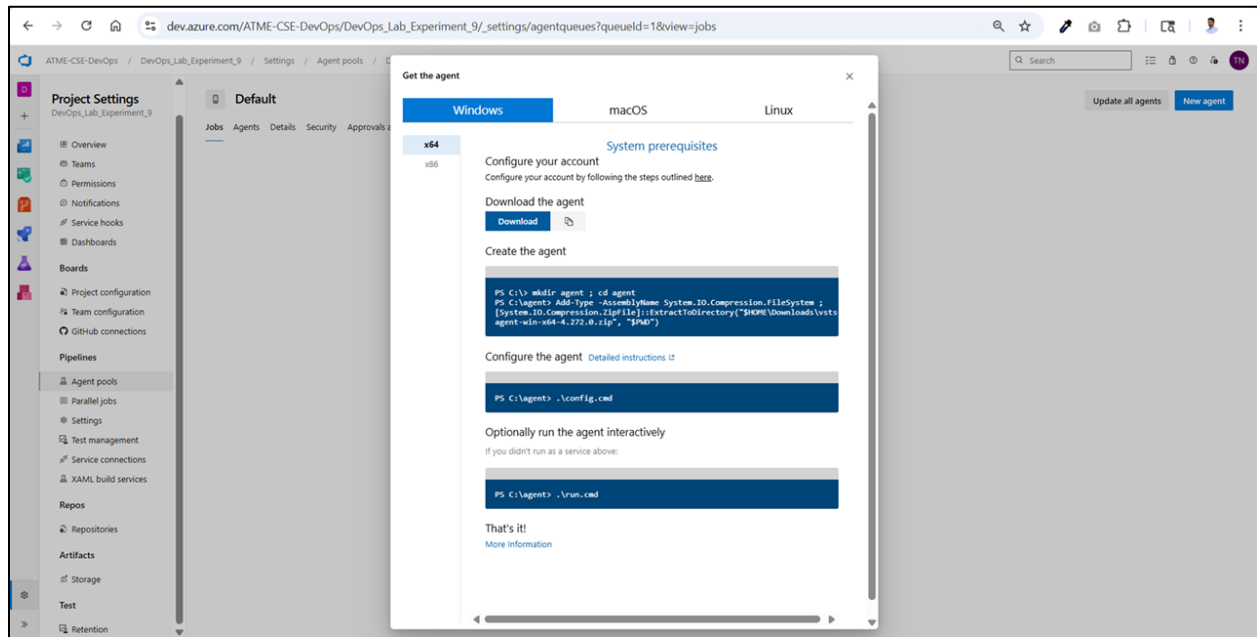
- Select **Project settings** → Under **Pipelines**, select **Agent pools** → **Default** → **New agent**.



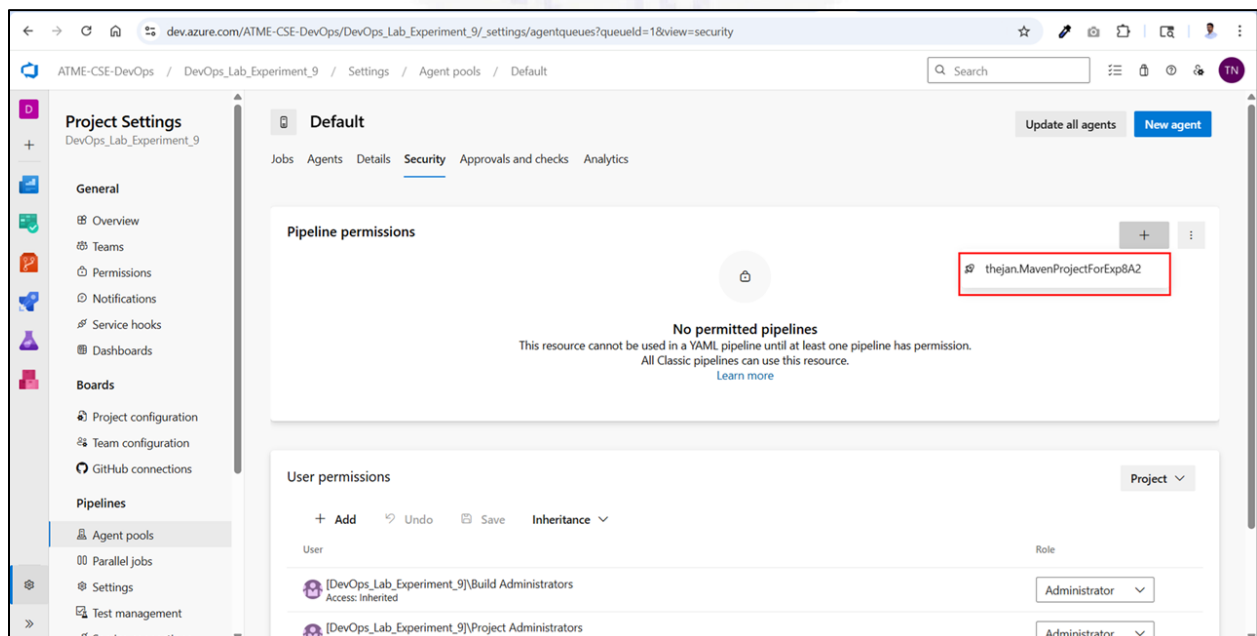
The screenshot shows the Azure DevOps project overview for "DevOps_Lab_Experiment_9". The left sidebar shows the "Project settings" menu. The main content area includes "About this project", "Project stats", and "Members". The "Project settings" menu item is highlighted with a red box.

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Create a directory named **Azure** in the C:\ drive, download the newly created agent into this directory, and then extract the zip file.

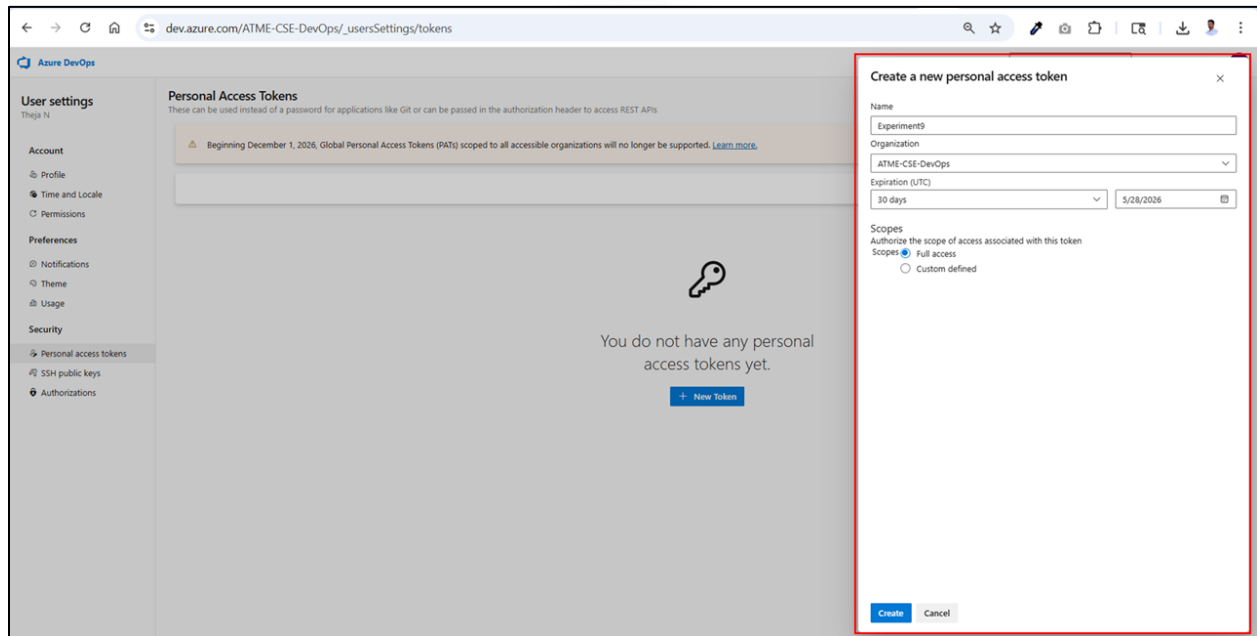


- **Configure Agent Pool Permissions :** Navigate to Project Settings → Agent Pools → Default → Security → Pipeline permissions. Click on the '+' (Add) button and select the required pipeline. Save the changes to authorize the pipeline to use the agent pool.



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- Open the **Windows PowerShell** (Run as administrator) and navigate to the agent directory.
- Configure the agent by `.\config.cmd`
- Give the organization URL for the server URL (Ex : <https://dev.azure.com/ATME-CSE-DevOps>)
- Go to the project dashboard → User settings → Personal access tokens → New Token. Provide Name, Select the Organization and Full access scope. Copy the new generated token.



- Go the Windows PowerShell and paste the token. Complete the configuration. After the configuration of the agent, the agent will be initially offline.
- Execute `.\run.cmd` to start the agent. The agent then connects to Azure DevOps Server; it becomes online and begins listening for jobs.

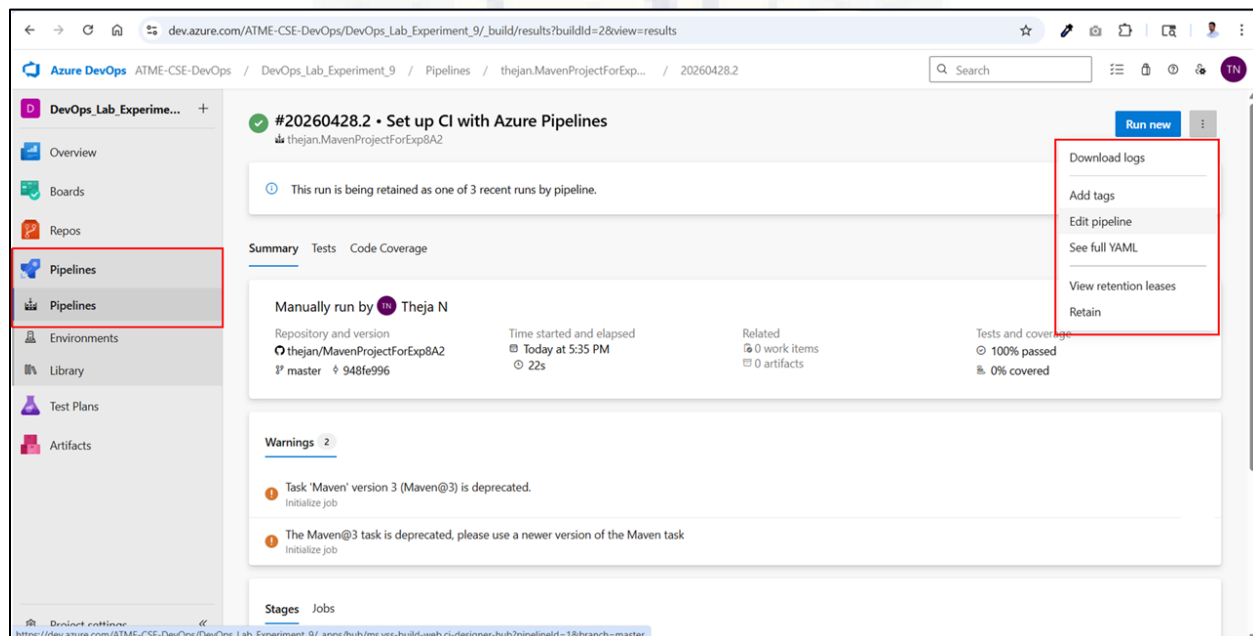
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```
Administrator: Windows PowerShell
PS C:\Azure\vsts-agent-win-x64-4.272.0> .\config.cmd

Agent Foundation
agent v4.272.0 (commit 491e611)

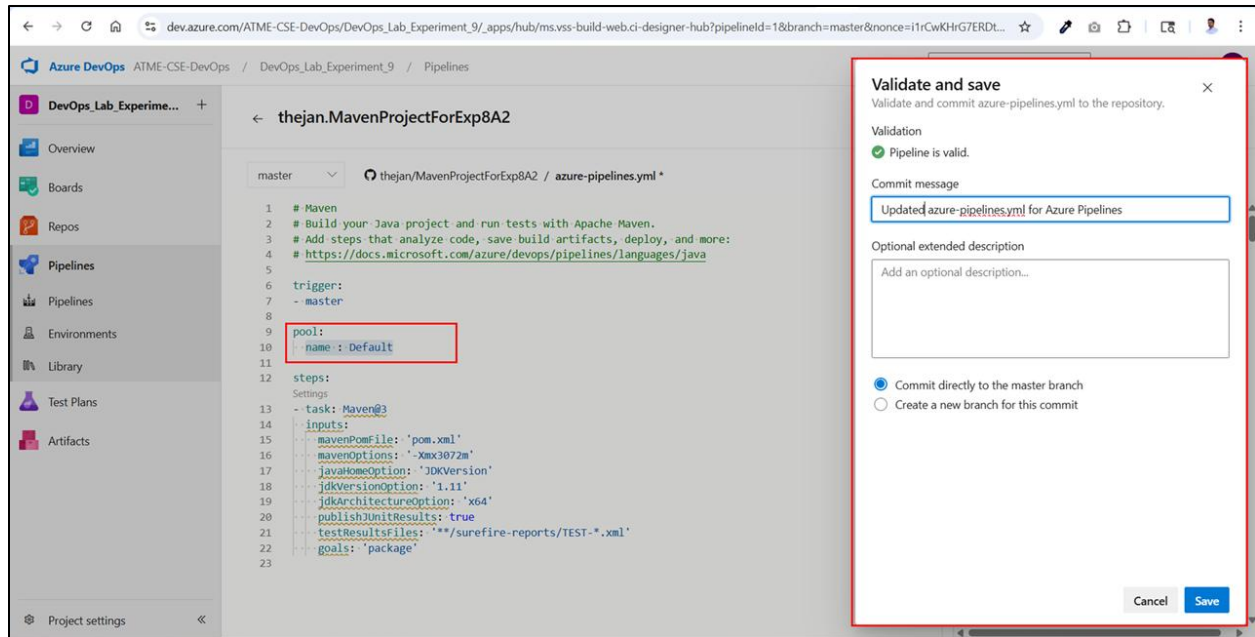
>> Connect:
Enter server URL > https://dev.azure.com/ATME-CSE-DevOps/
Enter authentication type (press enter for PAT) >
Enter personal access token > *****
```

Step 8 : Edit the pipeline to run the job on the self-hosted agent on the local machine.

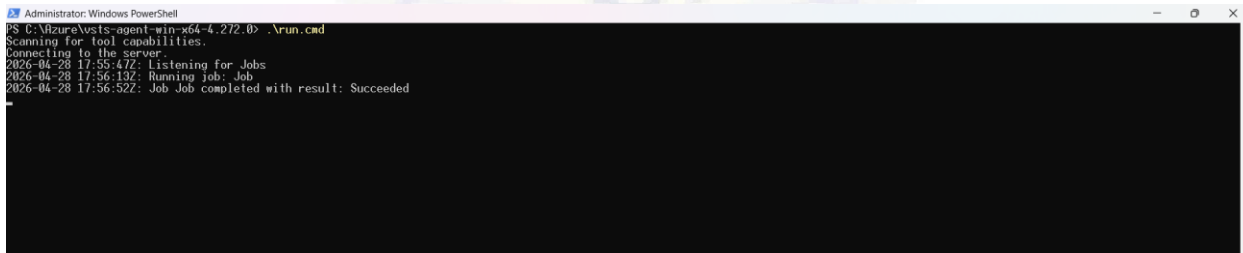


- Change the pool name to **Default** → **Validate and save**

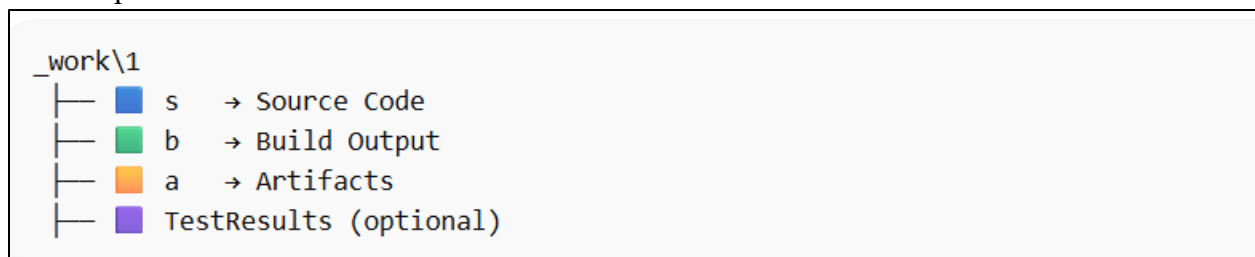
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- Go Pipelines → Select the pipeline → Click **Run new**
- This time, job on the local machine agent.



- You can verify the output of the job executed by the agent in the `_work` directory for a specific build.



_work Directory of Agent – Build Pipeline Structure

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VIVA QUESTIONS

1. What is Azure DevOps?
2. What is the purpose of using Azure Pipelines?
3. What is CI/CD in DevOps?
4. What is a pipeline in Azure DevOps?
5. What is the role of the azure-pipelines.yml file?
6. Why do we connect GitHub to Azure DevOps?
7. What is a repository?
8. What is an agent in Azure DevOps?
9. What is a self-hosted agent?
10. Why do we use a self-hosted agent instead of a Microsoft-hosted agent?
11. What is an agent pool?
12. What is the purpose of installing an agent on a local machine?
13. What is a Personal Access Token (PAT)?
14. Why is PAT required during agent configuration?
15. What happens when we execute config.cmd?
16. What is the purpose of run.cmd?
17. What does “agent is listening for jobs” mean?
18. What happens when a pipeline is triggered?
19. What is a build process in CI?
20. What is an artifact in Azure DevOps?
21. Where are artifacts stored after a build?
22. What is the difference between build pipeline and release pipeline?
23. What is a release pipeline?
24. What is deployment in DevOps?
25. What is the use of Command Line task in release pipeline?
26. How do you define variables in YAML?
27. How do you use variables in a pipeline?

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28. Where can you see the output of a pipeline execution?
29. What happens if the agent is offline?
30. What is the working directory of an agent?
31. What is the purpose of the _work folder?
32. What is GitHub integration in Azure DevOps?
33. What permissions are required for an agent pool?
34. What is meant by pipeline permissions?
35. What happens if no steps are defined in a stage?
36. What is meant by pipeline logs?
37. How do you rerun a failed pipeline?
38. What is the purpose of echo command in pipelines?
39. What are common errors in pipeline execution?